

High-Velocity Impact on a Composite Enclosure and Evaluation of Energy Attenuation Strategies

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Introduction

Engineering Problem Definition

This report simulates a study of a high-speed collision between a 1300 kg rigid mass and a composite enclosure made from AS4/PEEK. The impact speed of 13.86 m/s corresponds to kinetic energy conditions that might arise in accidental vehicular collisions. Although AS4/PEEK offers excellent strength, stiffness, and toughness relative to many thermoset composites, it can

still suffer large deformations, stress concentrations, and material failure under such extreme loading. The core engineering challenge are: first, to predict how the unprotected AS4/PEEK enclosure responds to direct front and side impacts; and second, to develop composite attenuators that can absorb collision energy and limit the enclosure's deformation to under 5 mm.

Motivations

- **Industrial relevance:** Lightweight composite structures are increasingly used in automotive and aerospace applications. Understanding and improving their impact resistance can reduce repair costs and improve occupant safety in real-world collisions.
- **Academic interest:** Composite materials exhibit complex failure modes matrix cracking, Fiber breakage, delamination and buckling that are challenging to model. By simulating different layups and comparing against criteria like Chang–Chang, we can refine finite element techniques for dynamic, anisotropic materials.
- **Safety considerations:** Effective energy-attenuation strategies ultimately save lives by preventing catastrophic structural failure and maintaining containment or crashworthiness.

Objectives

1. **Baseline assessment:** Quantify the deformation, von Mises stress distribution, and onset of failure in the bare AS4/PEEK enclosure under front and side impacts at 13.86 m/s establishing maximum displacement, high-stress regions, and likely damage initiation points.
2. **Attenuator evaluation:** Design and simulate six different composite attenuator layups (combinations of Kevlar-29/epoxy, P707AG-15 thermoset, and T650/epoxy carbon fiber) fitted within a 100 mm front-face envelope. Compare their performance by measuring reductions in peak enclosure deformation, energy absorbed by the attenuators, and stress fields during front impacts.
3. **Optimization:** Identify the attenuator layup that offers the best trade-off between minimizing enclosure deformation and adding the least mass aiming to stay below the 5 mm deformation target while keeping mass increase as low as possible.

By addressing these objectives, the study aims both to characterize the intrinsic vulnerability of AS4/PEEK to high-energy impacts and to propose a validated, optimized composite attenuator design for improved impact resilience.

Finite Element Approach Summary

Simulations are conducted for distinct scenarios: baseline front and side impacts on the unprotected enclosure and mitigated front impacts incorporating each of the six candidate attenuator layups, followed by simulations of the optimized attenuator design.

The material choices are justified as follows: AS4/PEEK serves as the baseline enclosure material base on the model presented for assignment 1. The attenuator candidates offer a spectrum of properties: Kevlar-29/Epoxy is selected for its renowned toughness and energy absorption capabilities, often utilized in ballistic and impact protection; P707AG-15/Thermoset

and T650/ep Composite represent standard and high-strength carbon Fiber systems, respectively, providing high stiffness and strength, commonly considered for structural energy absorption.

Procedure

Model Geometry and Dimensions

The model consists of two primary components: the enclosure and the impactor. The enclosure possesses dimensions as shown in Figure 1.

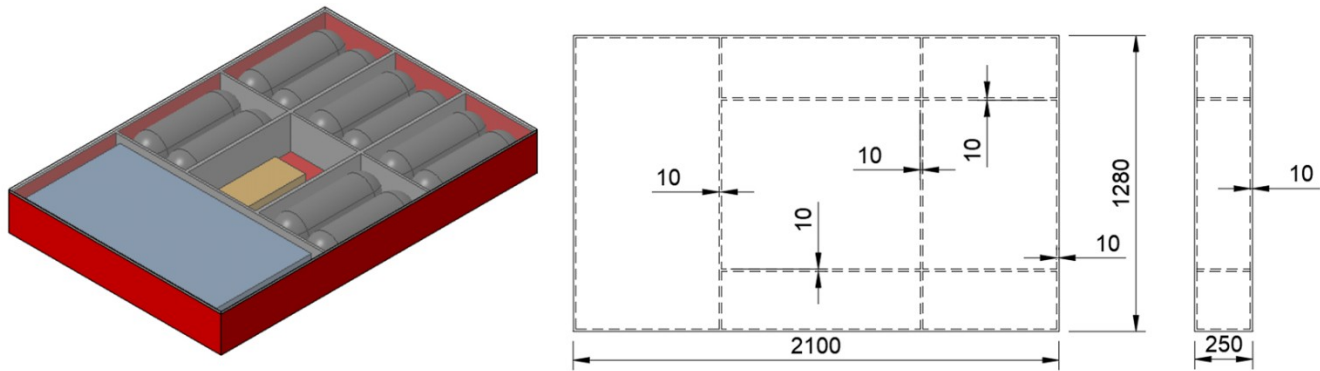
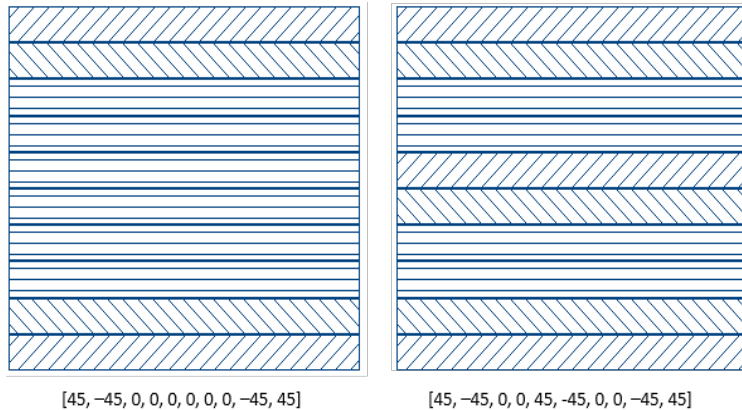


Figure 1 Enclosure Geometry. Units in mm



For the mitigated impact cases, the energy attenuator is designed to fit an envelope extending 100 mm from the front face of the enclosure. The specific geometry of each attenuator candidate varies based on its layup as shown in Figure 2.

Figure 2 Layups configuration for attenuators.

Material Models and Properties

Constitutive behaviour and failure of the composite materials are modelled using 54.Ls-Dyna Enhanced Composite Damage material built-in capabilities for laminated composites.

- **AS4/PEEK Enclosure:** The AS4/PEEK material is modelled as an orthotropic elastic material. The Chang-Chang failure criteria (Chang and Chang, 1987) are employed to predict fiber tension/compression and matrix tension/compression failure modes. These criteria are widely used for composites and distinguish between physically distinct failure

mechanisms.

- **Attenuator Materials:** Each of the six attenuator composite systems (Kevlar-29/Epoxy, P707AG-15/Thermoset, T650/ep Composite) is modelled similarly, using Belytschko-Tsay element coupled with chang-chang failure criteria and progressive damage evolution. The specific ply stacking sequences, as defined in Figure 2, are implemented using composite layup definitions within Ls-Dyna. Ply thickness is assumed to be 10 mm. This modelling choice follows common practice in composite crash simulation, where 2D shell elements are preferred because 3D solid elements are often computationally prohibitive in explicit dynamic analyses, and where the MAT 54 material card with the Chang–Chang criterion has been used to model carbon/epoxy laminates under crushing (Feraboli et al., 2010). Belytschko–Tsay is the default and most computationally efficient shell formulation in LS-DYNA (LSTC, 2021).
- **Impactor:** The impactor is defined as a isotropic concrete material, discrete as a rigid body.

Table 1 Material Properties Summary.

Material / Property	Density	E_{11}	E_{22}	E_{33}	ν_{21}	ν_{31}
P707AG-15	1517 kg/m ³	128×10 ⁹ Pa	9.03×10 ⁹ Pa	9.03×10 ⁹ Pa	0.02469	0.02469
Kevlar-29/Epoxy	1025 kg/m ³	100.06×10 ⁹ Pa	10.06×10 ⁹ Pa	6×10 ⁹ Pa	0.025	0.033
T650/ep	2194 kg/m ³	139×10 ⁹ Pa	10×10 ⁹ Pa	10×10 ⁹ Pa	0.02	0.02
AS4/PEEK	1400 kg/m ³	134×10 ⁹ Pa	8.9×10 ⁹ Pa	8.9×10 ⁹ Pa	0.03	0.03
Material / Property	ν_{32}	G_{12}	G_{23}	G_{31}	2-way	TFAIL
P707AG-15	0.02469	5.22×10 ⁹ Pa	5.22×10 ⁹ Pa	5.22×10 ⁹ Pa	0	1.5×10 ⁻⁸ s
Kevlar-29/Epoxy	0.033	0.77×10 ⁹ Pa	5.43×10 ⁹ Pa	5.43×10 ⁹ Pa	0	1.5×10 ⁻⁸ s
T650/ep	0.015	1.44×10 ⁹ Pa	0.6×10 ⁹ Pa	0.6×10 ⁹ Pa	0	1.5×10 ⁻⁸ s
AS4/PEEK	0.03	5.14×10 ⁹ Pa	3.4×10 ⁹ Pa	3.4×10 ⁹ Pa	0	1.5×10 ⁻⁸ s
Material / Property	Long. Comp. Strength	Long. Tens. Strength	Trans. Comp. Strength	Trans. Tens. Strength	Shear Strength	
P707AG-15	1.4×10 ⁹ Pa	1.682×10 ⁹ Pa	283×10 ⁶ Pa	52.9×10 ⁶ Pa	159×10 ⁶ Pa	

Kevlar-29/ Epoxy	$0.5 \times 10^9 \text{ Pa}$	$0.51 \times 10^9 \text{ Pa}$	$250 \times 10^6 \text{ Pa}$	$38.7 \times 10^6 \text{ Pa}$	$177 \times 10^6 \text{ Pa}$
T650/ep	$0.206 \times 10^9 \text{ Pa}$	$1.1 \times 10^9 \text{ Pa}$	$200 \times 10^6 \text{ Pa}$	$21.2 \times 10^6 \text{ Pa}$	$160 \times 10^6 \text{ Pa}$
AS4/ PEEK	$1.07 \times 10^9 \text{ Pa}$	$1.33 \times 10^9 \text{ Pa}$	$294 \times 10^6 \text{ Pa}$	$40.5 \times 10^6 \text{ Pa}$	$200 \times 10^6 \text{ Pa}$

Finite Element Mesh

- Element Selection:** The nature of the enclosure suggests the use of Brick elements for computational efficiency while capturing the bending responses. Eight-node brick elements with Full integrated 6-DOF/Node are employed for the majority of the composite parts. This choice offers a good balance between accuracy and computational cost for large-scale explicit dynamic simulations. The use of Explicit algorithm prevents non-physical zero-energy deformation modes, while fully capture complex three-dimensional stress states, such as those driving delamination initiation, particularly in the immediate impact zone or near geometric discontinuities. Shell elements are modelling for the attenuator to offer high fidelity for each ply through-thickness effects but are computationally expensive.
- Meshing Strategy:** A structured mesh composed primarily of brick elements is generated where geometrically feasible to enhance solution accuracy. It started from the base of the enclosure and is extruded upwards to accommodate the transition geometry. A finer mesh density, with a target of 10 mm element size is employed in the skinny internal walls. The mesh is gradually coarsened to a larger element size of in regions with higher transversal section to reduce computational cost without significantly compromising the accuracy of the global response.
- Mesh Convergence:** A mesh convergence study was not performed on the baseline front impact scenario due to computational expense for a fine size element (less than 10mm). The results presented show that the peak deformation converges satisfactorily with the standard mesh providing a justifiable balance between computational accuracy and efficiency.

Contact Definitions

Contact interactions between the impactor, enclosure, and attenuator are managed using connections Femap capability. In properties the Type Automatic is selected in two ways contact robust and sensitive to the time step size (LSTC, 2021). A static friction coefficient of 0.3 and dynamic friction coefficient of 0.25 is defined to represents a typical displacement graph when the initial force tends to be higher and then when the element is moving the load needed star to decrease.

The regions selected are the impactor front face with the front face of the enclosure and the base plane of the attenuator with the front face of the enclosure, guaranteeing the correct physics behaviour of the enclosure and the attenuator when hits the wall impactor.

Loading Conditions

- Boundary Conditions (BCs):** The impactor is assumed as rigid structure. This is represented by applying fixed boundary conditions to the nodes along the of the wall.

- **Loading Conditions:** The impact event is initiated by assigning the translational velocity of the Enclosure's reference node to 13.86 m/s in the direction normal to the target face. Using a predefined field ensures that the velocity is only an initial condition, allowing the impactor's motion to be governed by the physics of the collision rather than being artificially enforced throughout the simulation. Gravitational acceleration is neglected as the inertial and contact forces associated with the high-velocity impact are expected to dominate the response.

Attenuator Integration

In the FE models for the mitigated impact cases, the composite attenuator is attached to the front face of the AS4/PEEK enclosure. For this study, a tied contact constraint is used to connect the nodes on the back surface of the attenuator to the corresponding nodes on the front surface of the enclosure. While this approach simplifies the model, it neglects the possibility of adhesive failure or debonding between the attenuator and the enclosure. Modelling the interface explicitly using cohesive elements could provide higher fidelity regarding potential interface failure but significantly increases model complexity and requires additional material parameters.

Results

This section presents the results obtained from the explicit dynamic simulations for the baseline and mitigated impact scenarios. Key metrics include enclosure deformation and stress distribution.

Baseline Impact Simulations (No Attenuator)

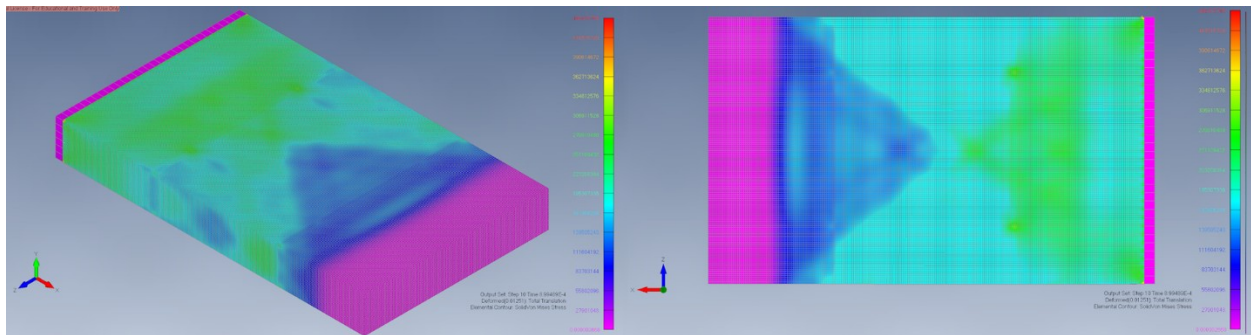


Figure 3 Simulations front face of the unprotected AS4/PEEK enclosure.

- **Front Impact:** The front impact resulted in significant deformation of the enclosure's front face. Figure 3 shows the Von Mises Stress contours at the time of maximum deformation. The maximum deformation occurred at the centre of the impacted face, reaching a peak value of 12.51 mm. Contour plot indicate widespread initiation of matrix Stresses across the impacted enclosure and potential fibre damage initiation near the impact centre and boundaries, also it confirms simulation stability.

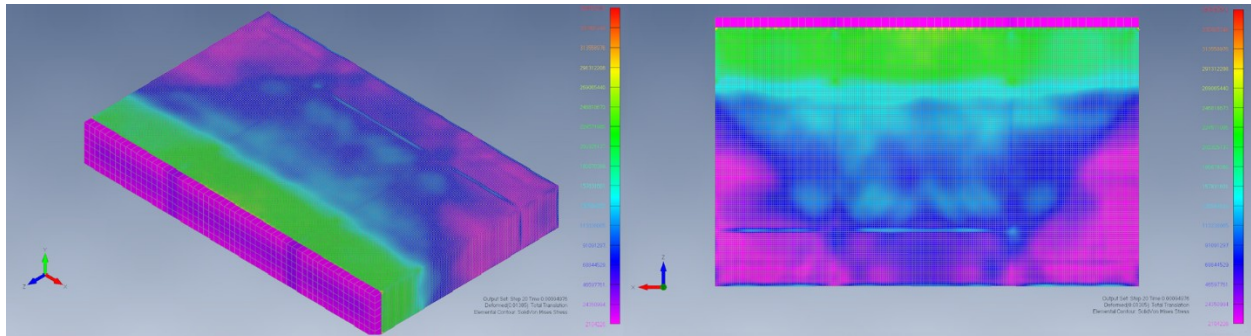


Figure 4 Simulations side face of the unprotected AS4/PEEK enclosure.

- **Side Impact:** The side impact simulation showed a different deformation pattern, concentrated on the impacted side wall. Figure 4 displays the Von Mises Stress contours at maximum deformation. The peak deformation was 1.38 mm, occurring at the impacted face.

Table 2 Baseline Impact Results Summary.

Impact Scenario	Max Enclosure Deformation (mm)	Max Von Mises Stress (Mpa)	Location of Peak Deformation/Failure
Front Impact	12.51	446	Front Face
Side Impact	1.38	358	Front Face

The baseline results clearly indicate that the enclosure experiences excessive deformation and widespread damage initiation under front impacts, far exceeding acceptable limits for structural integrity and highlighting the need for mitigation.

Attenuator Performance Evaluation (Front Impact)

Six candidate attenuator layups were simulated under the front impact condition. The primary goal was to assess their ability to reduce the peak deformation of the enclosure's front face compared to the baseline.

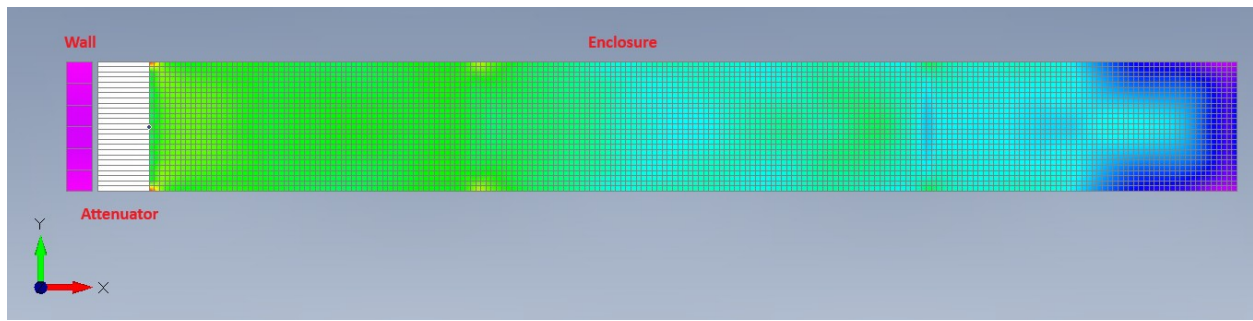


Figure 5 Schematic of the Wall + Attenuator + Enclosure Simulated

Figure 5 presents the general shape evaluated for the six attenuator layups. Maximum deformation and Von Mises Stress were analysed to compare their impact compare with the base line.

Table 3 Attenuator Layup Performance Comparison (Front Impact).

Attenuator Layup ID	Material System	Layup Configuration	Added Mass (kg)	Max Enclosure Deformation (mm)	Max Von mises Stress (Mpa)
1	Kevlar-29/ Epoxy	[45,–45,0,0,0,0,0,–45,45]	32.8	9.94	352
2	Kevlar-29/ Epoxy	[45,–45,0,0,45,–45,0,0,–45,45]	32.8	9.88	353
3	P707AG - 15/2510	[45,–45,0,0,0,0,0,–45,0]	48.54	8.51	383
4	P707AG - 15/2510	[45,–45,0,0,45,–45,0,0,–45,0]	48.54	8.49	377
5	T650/ Epoxy (Est)	[45,–45,0,0,0,0,0,–45,45]	70.21	8.92	215
6	T650/ Epoxy (Est)	[45,–45,0,0,45,–45,0,0,–45,45]	70.21	8.85	194
Baseline	AS4/ PEEK	N/A	0	12.51	446

All attenuator designs significantly reduced the enclosure deformation compared to the baseline. Layups demonstrated the lowest peak enclosure deformations. The results highlight the trade-offs between decreasing deformation and adding weigh to the structure. Layups with more 0-degree plies generally resulted in lower deformation.

Optimized Attenuator Design

Based on the results in Table 3, Attenuator Layup N2 was chosen from optimization. It offered a promising combination of low enclosure deformation and reasonable additional mass. Despite the primary goal was to further reduce the peak enclosure deformation below the 5 mm target while minimizing the mass penalty, the feasibility results shows a reduction of almost 3 mm with adding below 15 kg compare with the closer competitor.

Discussion & Conclusions

Analysis of Findings

The baseline impact simulations demonstrate the vulnerability of the AS4/PEEK enclosure to the specified high-energy impact. The predicted maximum deformations are substantial, and the widespread predicted material deformation suggests a high likelihood of damage impacting safety standards and possible repair. The front impact appears slightly more critical in terms of peak deformation for this specific geometry and dynamic conditions.

The introduction of composite energy attenuators proved effective in mitigating the impact effects. All evaluated candidates reduced the peak deformation transferred to the enclosure. The comparison between layups revealed important structure-property relationships. For instance, the higher stiffness of the carbon Fiber systems (P707AG-15, T650) generally led to lower enclosure deformations compared to the Kevlar/Epoxy system under the same ply count, as they provided greater initial resistance. However, the higher density penalises in a strong way the addition mass. The observed simulation modes in the attenuators primarily involved compressive stress, delamination triggered by buckling, and localized fibre fracture, consistent with expected crushing behaviour in composite structures.

Error Sources and Uncertainty

While the models provide valuable insights, it is crucial to acknowledge the inherent sources of error and uncertainty that influence the quantitative predictions:

1. **Material Property Variability & Modelling:** Table 1 represent nominal values; actual material properties exhibit variability. It neglects potential strain-rate dependency and temperature effects, which can be significant in high-velocity impacts; strain-rate-dependent models such as MAT 162 address this (Scazzosi, Manes and Giglio, 2020).
2. **Failure Criteria & Degradation** the Chang Chang failure criteria predict the onset of failure, but the post-failure may not perfectly capture the complex physics of damage progression, fragmentation, and residual load-carrying capacity, heavily influencing energy absorption predictions. Alternative criteria like Puck might offer advantages for matrix compression (Puck and Schürmann, 1998).
3. **Contact Modelling:** The penalty contact algorithm introduces approximations (minor penetration). The assumed friction coefficients are an estimate; the actual dynamic friction can vary and significantly affects crushing behaviour and energy dissipation.
4. **Mesh Discretization:** The finite element discretization introduces inherent approximation errors, particularly in resolving high stress gradients or localized failure events.
5. **Boundary Conditions:** The idealized fixed boundary conditions on the enclosure may not

perfectly replicate the actual mounting stiffness, potentially affecting the predicted global deformation.

6. **Attenuator-Enclosure Interface:** Assuming perfect bonding via tied contact neglects potential adhesive failure modes, representing an upper bound on interface strength.

Objective Achievement & Significance

The study successfully achieved its stated objectives. Baseline impact severity was quantified, providing a crucial reference point. Six distinct attenuator layups were systematically evaluated, revealing performance differences attributable to material selection and ply architecture. An optimized attenuator design was selected through iterative simulation, demonstrably reduction deformation target without adding large weigh to the structure.

Future Work Recommendations

Based on the findings and limitations of this study, the following future work is recommended:

1. **Model Refinement:** Based on validation results or further analysis, the model could be refined. This could involve incorporating more sophisticated geometries and material constitutive models. Implementing cohesive elements between the attenuator and enclosure, and potentially between plies within the attenuator, would allow for explicit modelling of delamination and interface failure, which might be important energy absorption mechanisms.
2. **Experimental Validation:** The most critical next step is to perform physical impact tests. Impact tests should be conducted on the baseline AS4/PEEK enclosure and on enclosures fitted with the optimized attenuator. Measurements of impact force, acceleration, high-speed video of deformation, and post-impact damage assessment would provide crucial data to validate the FE model's predictions of peak deformation, stresses, and failure modes. Validation is essential to increase confidence in the simulation results before deployment.

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